Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details

a. Team name: Agamya
b. Team leader name: Siddhi Pandirkar
c. Problem Statement: Future of Work & Enterprise Productivity

Brief about the idea

When an expert leaves, the *why* leaves with them. It has happened for centuries.

Bengal, muslin

One of history's finest cloths, spun from phuti karpas cotton. The cotton and the skills are gone.

The "cut thumbs" story is legend; the record shows tariffs and machine-made imports. [1]

Sikkim, Majhi (माझी)

2013: India's last known speaker, Thak Bahadur Majhi, aged 80. Majhi has 15 words for river.

Still spoken in Nepal, where Nepali is replacing it. [2]

Japan, traditional crafts

Registered craft workers: about 288,000 (1979) to 48,334 (FY2022).

In Kanazawa, craft families close for lack of successors. [3]

Every company

42% of institutional knowledge is unique to the employee in their role.

Panopto survey, 2018. [4]

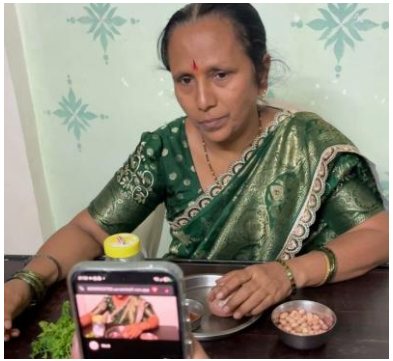
Agamya captures the why, live, and passes it on

1 The expert works as usual, on screen or camera. **Moti**, a Gemini Live agent, asks one question at a pause.

2 A Work Map: steps, decisions, reasons in the expert's own verified words, and guardrails.

3 Learners practise with Moti as tutor, in their language. AI agents ask "would the expert stop here?"





Real sessions: a JMeter handover before leave · a Satara recipe in Marathi

[1] Daily Star 2026; Wikipedia [2] Down To Earth 2013; Wikipedia [3] Garland 2026, citing Assoc. for the Promotion of Traditional Craft Industries; NPR 2022 [4] Panopto 2018. Full references: slide 14 and notes.

Opportunities

How different is it from any of the other existing ideas?

Others	Agamya
Screen recorders, SOP writers record what was clicked	**ask why, live, at a pause**
Document chatbots may invent answers	**the expert's verified quote, or "AI-inferred"**
Training videos watch and forget	**practise; a tutor stops mistakes**
Agent guardrails hand-written prompts	**from the expert, machine-checkable**
English-first tools	**Marathi, Hindi, Kannada and more**

How will it be able to solve the problem?

1 Capture
Moti asks why while the expert works

↓

2 Understand
Work Map with verified quotes

↓

3 Confirm
the expert approves and chooses who learns

↓

4 Transfer
practice, listening, Ask, AI agents

↓

5 Measure
impact from records, not estimates

Minutes of the expert's time.

USP of the proposed solution

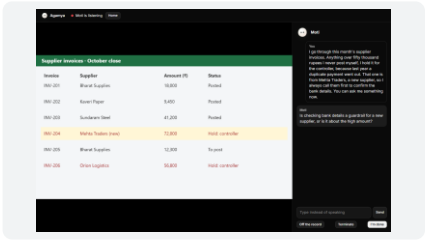
One confirmed judgment, in the expert's own words, becomes three things:

a lesson
for people, in 15 languages

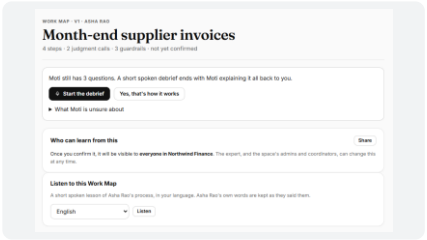
a live tutor
that stops a wrong decision

a guardrail
for AI agents (REST, MCP, ADK)

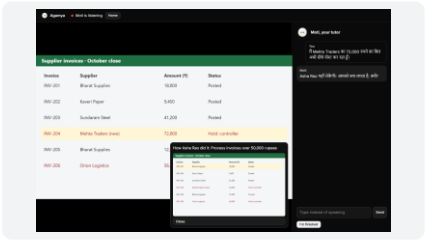
List of features offered by the solution



1. Live capture with Moti
Gemini Live asks at a pause; off the record any time



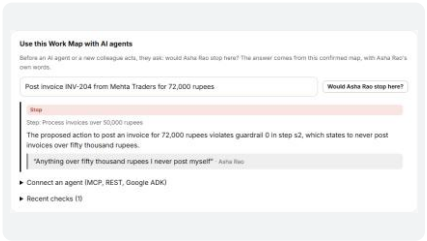
2. Work Map
steps, reasons in the expert's verified words, guardrails



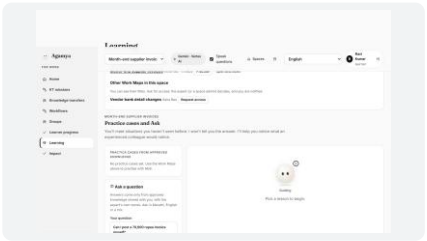
3. Practise with a tutor
camera or screen; Moti stops a wrong decision



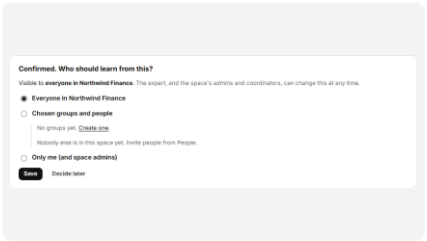
4. Listen
a spoken lesson in 15 languages



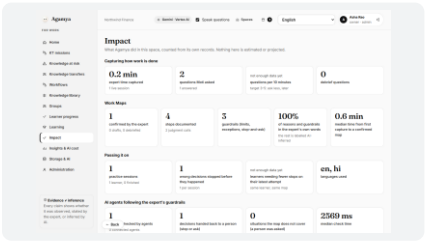
5. AI Guardian
STOP / ASK / PROCEED via REST, MCP, ADK



6. Ask
answers only from shared maps, quoting the expert



7. Confirm and share
space, groups or people; request access



8. Impact, measured
minutes, maps, mistakes stopped, AI cost

Also: debrief and teach-back · spaces and roles · optional space passcode · consent, masking of personal data, per-space isolation · works on phones (camera)

Process flow diagram or Use-case diagram

Expert works
screen or camera

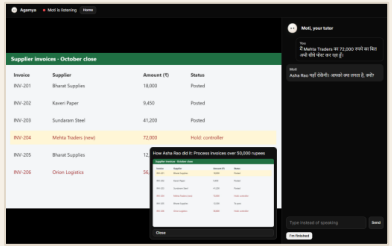
→ **Moti asks why**
one question at a pause

→ **Work Map**
verified quotes, guardrails

→ **Debrief**
Moti explains it back

→ **Expert confirms**
and chooses who learns

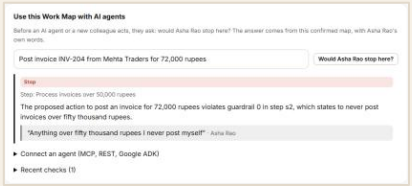
Learners practise
Moti stops mistakes, in their language



Listen and Ask
15 languages; answers quote the expert



AI agents check
"Would the expert stop here?"

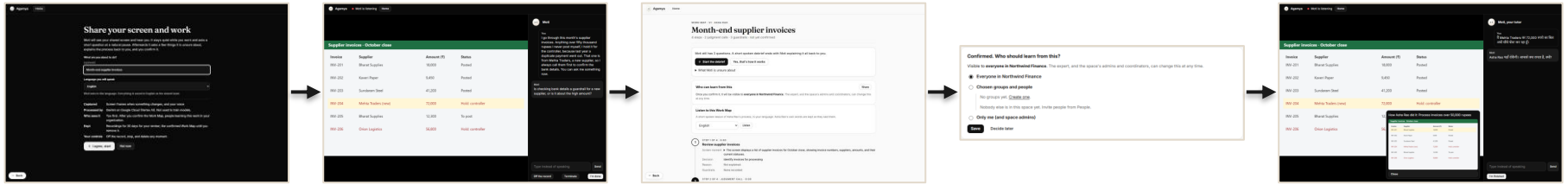


Work: accounts payable, IT runbooks, escalations, a retiring specialist

Impact, measured

Culture: a Kathak guru's footwork, a weaver's loom, a family recipe in Marathi

Wireframes/Mock diagrams of the proposed solution (optional)



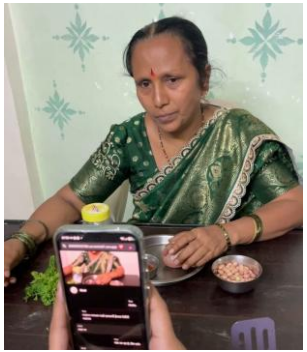
1 Consent
what is captured, who sees it

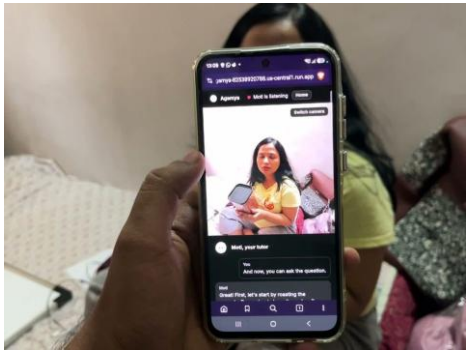
2 Moti asks
at a pause, by voice

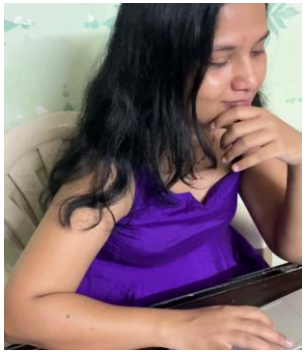
3 Work Map
steps, quotes, guardrails

4 Confirm, share
who learns from this

5 Practise
Moti stops a mistake







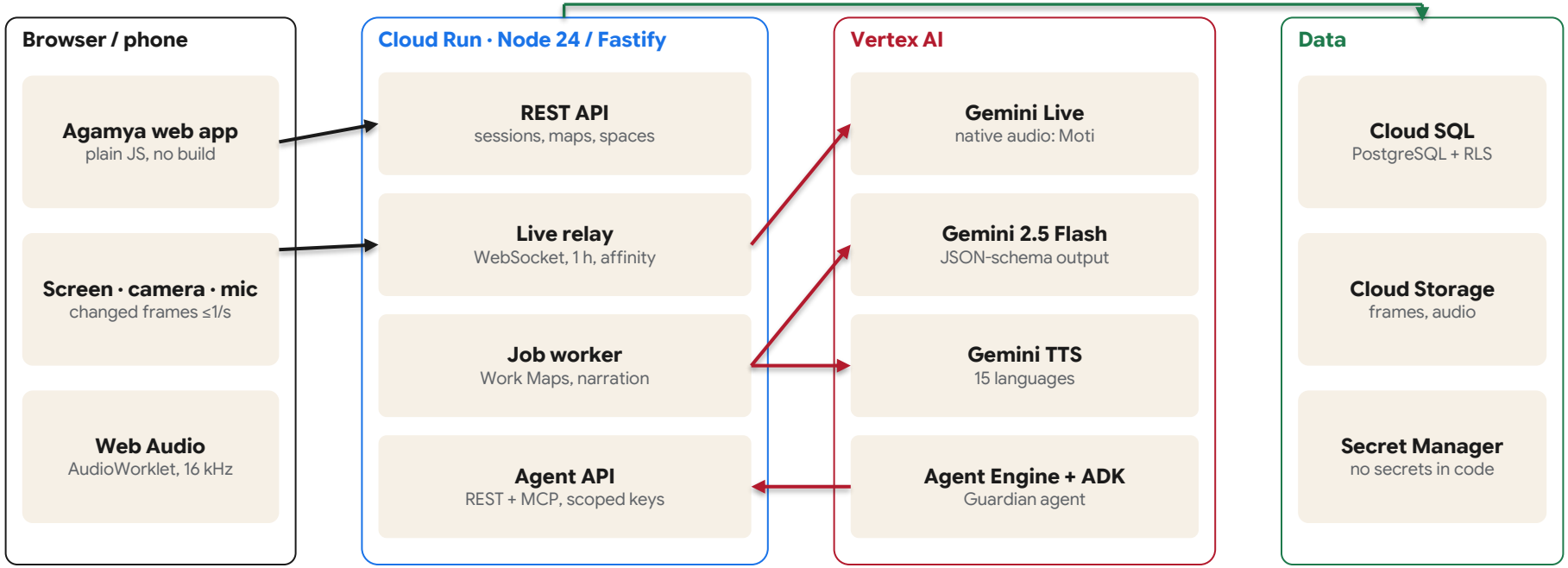
In real use: Bharati records with Moti · Kiran practises with Moti · an expert shares her screen

Not mock-ups

The top row is the deployed prototype. The photos are real sessions recorded on phones: a family recipe in Marathi, and a work handover on a laptop.

Tested on iPhone and Android.

Architecture diagram of the proposed solution



Flow: audio and changed frames go through the relay to Gemini Live, and Moti's voice comes back. The worker builds Work Maps with Gemini 2.5 Flash; every quote is checked in code. External AI agents, and the ADK Guardian on Agent Engine, call the Agent API before acting. Row-Level Security isolates each space in the database.

Technologies to be used in the solution

Google Cloud, and why

Gemini Live API gemini-live-2.5-flash-native-audio	Moti: real voice conversation with vision, in Indian languages
Gemini 2.5 Flash structured JSON output	Work Maps, tutor stops, agent checks, Ask: fast and cheap, checked in code
Gemini TTS gemini-2.5-flash-tts	narration in 15 languages
Vertex AI Agent Engine + ADK Python Guardian agent	managed agent hosting with tracing
Cloud Run WebSockets, 1 h timeout, affinity	the whole app; scales to zero
Cloud SQL · Storage · Secret Manager PostgreSQL 16 + RLS	isolation enforced by the database; private media; no secrets in code
Cloud Build · Artifact Registry Cloud Logging	one-command deploy: npm run deploy

Also used

Node.js 24 · Fastify
server and WebSocket relay

Plain JavaScript web app
no build step, works on phones

Web Audio · media capture
AudioWorklet, screen and camera

Model Context Protocol
agents call the guardrails as a tool

PGlite
local development database

axe-core · Playwright
accessibility checks, screenshots

Public datasets
YouCook2, ai4privacy (evaluation)

Estimated implementation cost (optional)

Build one Work Map (Gemini 2.5 Flash)	**≈ $0.005**	*Measured: 50 builds cost $0.23*
10-minute live capture with Moti	**≈ $0.10–0.20**	*Estimate at list prices*
Spoken lesson, per language (cached)	**≈ $0.01**	*Estimate at list prices*
One agent check	**< $0.001**	*Estimate at list prices*
Cloud Run	**pay per use**	*scales to zero*
Cloud SQL (db-f1-micro) + Storage	**≈ $10–15/mo**	*pilot size, estimate*

Example

One expert, 20 workflows, about 30 minutes each:

under $15 of AI

for a fully mapped role (estimate)

How we know: every AI call is logged with its list-price cost, and the Impact page shows AI cost per map. Green = measured; gold = estimate.

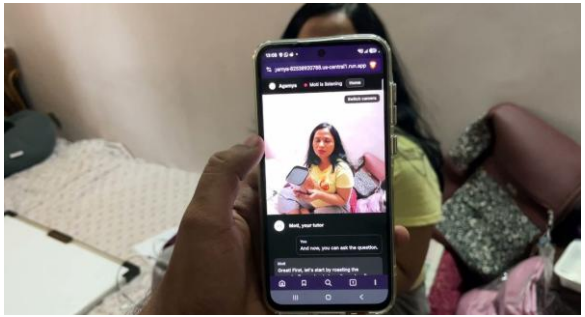
Snapshots of the prototype



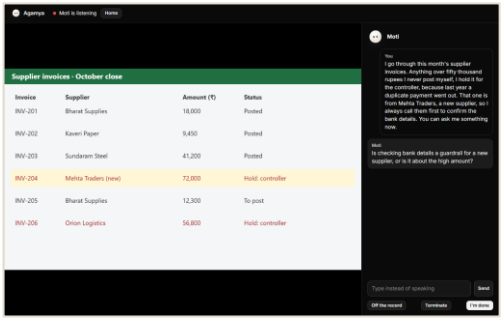
Work: an expert shares her screen



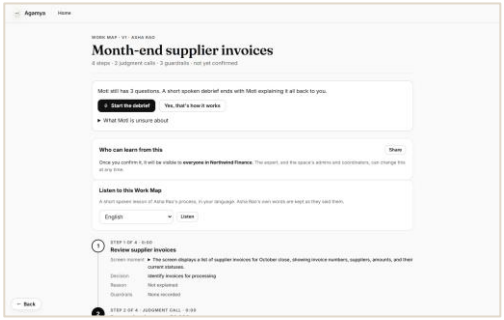
Culture: Moti asks in Marathi



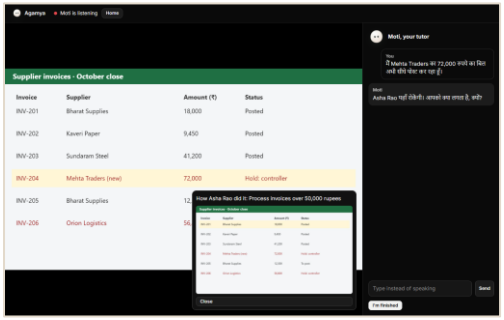
Kiran practises; Moti tutors



Moti asks a guardrail question



Work Map, verified quotes



Tutor stops a mistake (Hindi)

Prototype Performance report/Benchmarking

176 / 176	22 / 22	≈ 2 s	≈ 1.6 s
automated tests pass	end-to-end checks on the live app	Moti asks after the expert pauses	tutor stops a wrong decision

Work Maps on 25 real recipes (YouCook2, 212 steps)

100%	0%	100%
steps covered	invented evidence	steps in order

Quotes the expert didn't say: silent videos 1 claimed → 0 verified; narrated 3 → 3. Unspoken quotes are labelled AI-inferred, every time.

Cost: 50 Gemini calls, $0.23.

Guardian agent on Agent Engine: returned STOP for ₹88,000 and ₹64,500 postings over the expert's limit (live test).

Personal data masked before storage



ai4privacy, 43,501 sentences. Amounts and dates: 0% changed.

All figures measured by the team (see docs/EVALUATION.md). Latencies are from a small number of live sessions. Also tested on iPhone.

Additional Details/Future Development (if any)

Next 3 months	6 months	12 months
Two experts on one map: where they agree, differ, and why	Export guardrails to agent platforms and RPA tools	Culture archive: opt-in community collections of crafts and languages
Native-speaker review of all 15 narration languages	On-device masking of faces and screens before upload	Partnerships with craft associations and language surveys
Google Workspace sign-in and directory groups	Expert-shift alerts when a confirmed rule changes	Certified practice paths for regulated roles



Our Inspiration

The app has a story page: Bengal muslin, Majhi, Japan's craft families, and the expert at your desk, with sources. The same loop that keeps an invoice rule can keep a weaver's reason.

agamya-82538920788.us-central1.run.app/#/inspiration

Provide links to your:

GitHub Public Repository
github.com/S-D-P/Agamya-AI-Builder-Cup-26

Demo Video Link (3 Minutes)
ADD YOUTUBE LINK

Final Product Link
agamya-82538920788.us-central1.run.app

Judge sign-in

One click fills these in on the sign-in page.

Work: judge.work@agamya.example
Culture: judge.culture@agamya.example
Password: agamya-judges-2026

Start here: docs/JUDGES_GUIDE.md in the repository shows every feature, one screenshot at a time.

Use Chrome or Edge on a computer for screen sharing; on a phone, use the camera. Allow the microphone and wear headphones.

References

[1] The Daily Star (2026). What Bengal made and let others claim. thedailystar.net · Wikipedia: Dhaka muslin.

[2] Chakravartty, A. (6 Sep 2013). Survey tracks down only speaker of Majhi in India. Down To Earth · Wikipedia: Majhi language.

[3] Garland Magazine (2026). Kimono and lacquerware sales down up to 90 per cent as Japan's craft base narrows (data: Association for the Promotion of Traditional Craft Industries) · NPR/WFAE (29 Jul 2022). Japan's traditional crafts are struggling to survive the country's population decline.

[4] Panopto (2018). Workplace Knowledge and Productivity Report, via HR Dive.

[5] Polanyi, M. (1966). The Tacit Dimension.

[6] Zhou, L., Xu, C., Corso, J. (2018). YouCook2 dataset · ai4privacy. pii-masking-200k (Hugging Face).

Agamya's own results: docs/EVALUATION.md. Full URLs in these notes and in docs/submission/RESEARCH_LEDGER.md.

Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.

Thank you

